Integer Programming Approaches to the Quadratic Knapsack Problem

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Abstract

This project focuses on the binary quadratic knapsack problem (QKP), first introduced by Gallo et. al. Its formal definition is as follows: Assume that n items are available, where item j has a positive integer weight w_j. Additionally, we are given an $n \times n$ nonnegative integer matrix P $=\{p_{ij}\}$, where p_{jj} is the profit achieved if item j is selected and $p_{ij} + p_{ji}$ is a "synergy" profit achieved if both items i and j are selected for $i < j$.

Given the indicies $[n] := \{1, \ldots, n\}$ to denote the item set N, we introduce a binary variable x_j to indicate whether item j is selected. The QKP calls for selecting an item subset that maximizes the overall profit, while fulfilling the knapsack capacity (budget) constraint.

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1 Introduction

The quadratic knapsack problem (QKP) can be written as:

$$\begin{aligned} \max \quad & \sum_{i \in N} \sum_{j \in N} p_{ij} x_i x_j \\ \text{s.t.} \quad & \sum_{j \in N} \omega_j x_j \leq c \\ & x_j \in \{0, 1\}, j \in N \end{aligned} \tag{QKP}$$

Without loss of generality we assume that $\max_{j \in N} \omega_j \leq c < \sum_{j \in N} \omega_j$ and that the profit matrix is symmetric, i.e., $p_{ij} = p_{ji}$ for all $j > i$.

Note, some formulations write the objective as

$$\max \sum_{i=1}^{n} p_i x_i + \sum_{i=1}^{n} \sum_{\substack{j=1 \\ i \neq j}} P_{ij} x_i x_j$$

to make the synergy profit more explicit; however, this is covered in our formulation for the construction of P as described. In particular, our formulation has

$$p_{ii} \cdot x_i^\top x = p_i x_i,$$

and we can scale synergy profits to $\hat{p}_{ij} = p_{ij}/2$ so that

$$p_{ij} x_i x_j = \hat{p}_{ij} x_i x_j + \hat{p}_{ji} x_j x_i$$

for any pairs x_i, x_j selected in the knapsack (including the "double count" of singleton items). We prefer (QKP) for its compactness, and its alignment with our computational component in §4.

2 Literature Review

Here we review the current literature relevant to our project. Much work has been done in both the realms of nonlinear programming and continuous relaxations for the 0-1 knapsack problem. We took a look at some of the leading contributors to the origin of solution methodologies for the problem, and detailed improvements made by several practitioners since the creation of these foundations.

2.1 Surveys

Pisinger's survey [29] provided a comprehensive review on the development of QKP's upper bounds and their application in variable reduction, as well as some heuristics, approximation algorithms, valid inequalities, and branch and bound algorithms. The survey's main focus concerned the introduction of each upper bound to the field, and their incorporation into different solution techniques. The survey also provided a comparison of the efficiency of different upper bounds.

Atamtürk [1] provided a review of recent developments on valid inequalities for several types of knapsack polyhedron, and focused on cover and pack inequalities and their superadditive lifting. He also aimed to give a unified description of the inequalities for the knapsack polyhedron and highlights the connection among the inequalities. New results, like the definition of integer cover inequalities, and a description of a necessary and sufficient facet conditions were also presented in the paper.

2.2 Algorithmic Approaches

Chen and Hao presented an iterated responsive threshold search (IRTS) approach for solving the quadratic knapsack problem [13]. With a basis of three "neighborhoods", the algorithm combined a threshold-based exploration phase and a descent-based improvement phase with a dedicated perturbation strategy. After a large amount of experiments were performed on certain instances, the method performed very well compared to current state-of-the-art methods, and also discovers 41 improved lower bounds. The key components of IRTS were analyzed to display their importance to the performance of the algorithm.

Pisinger et al. discussed the solutions of large-sized QKPs through aggressive reduction [30]. In the paper, their main focus were the usable bounds for reduction of the size of the instance (to a manageable size). Specifically, the authors stated "the upper and lower bounding phases are repeated several times, each time improving the subgradient method used for finding the Lagrangian multipliers associated with the upper bounds. Having reduced the instance to a (hopefully) reasonable size, a branch-and-bound algorithm based on the Caprara-Pisinger-Toth framework is applied."

Fomeni and Letchford worked on a dynamic programming heuristic for QKP [16]. Their main purpose was to show that one can devise an effective heuristic using dynamic programming. The original pseudo code and its enhancements were introduced as the core part of the paper. They also conducted computational experiments to prove that classical dynamic programming approach to the standard (linear) knapsack problem can be modified to yield an effective heuristic for the quadratic version and the result was of high quality.

2.3 Existing Solution Methods

The QKP is usually solved by deriving good upper bounds and using approximation techniques until some tolerance parameter is satisfied, i.e. we have an integer (binary) solution that is "close enough" to the upper bound. Techniques to reduce the complexity of deriving these upper bounds include relaxation from upper planes, linearization, reformulation, Lagrangian relaxation, Lagrangian decomposition, and semidefinite programming [29]. Since any feasible integer solution, gained from heuristics or exploitation of the problem structure, gives lower bounds, solution techniques focus on closing the gap between the upper and lower bounds [30]. We will speak to several of these techniques here and several lower bounds. There are many upper bounds that have been created for the QKP during the last three decades. The survey from David and other papers (e.g. [16]) provided several important upper bounds, that we will summarize here. In total, we will introduce seven types of upper bounds for the QKP, with a brief discussion on their differences and efficiency.

2.4 Upper Bounds

2.4.1 Upper bounds from Gallo et al.

Gallo et al. first introduced the QKP and also presented the first bounds for QKP using the concept of upper planes [18]. The definition of an upper plane is a linear function f satisfying

$$f(x) \geq \sum_{i \in N} \sum_{j \in N} p_{ij} x_i x_j$$

for any feasible solution x of (QKP). In the paper, the authors utilize the form of $\sum_{j \in N} \pi_j x_j$ to represent the upper plane and the problem is reconstructed as the following relaxed optimization problem:

$$\begin{aligned} \max \quad & \sum_{j \in N} \pi_j x_j \\ \text{s.t.} \quad & \sum_{j \in N} \omega_j x_j \leq c \\ & x_j \in \{0,1\}, j \in N \end{aligned} \tag{1}$$

Gallo et al proposed 4 different upper bounds $U^i_{GHS}, i = \{1, \ldots, 4\}$, which are delineated by four different expressions of π_j:

$$\pi^1_j := \sum_{i \in N} p_{ij}$$

$$\pi^2_j := \max\{\sum_{i \in N} p_{ij} \bar{x}_i : \sum_{i \in N} \bar{x}_i \leq k,$$

where

$$k = \min\{h : \sum_{j=1}^{h} \omega_j > c\} - 1, \bar{x}_i \in \{0,1\} \text{ for } i \in N\}$$

$$\pi_j^3 := \max\{\sum_{i \in N} p_{ij}\bar{x}_i : \sum_{i \in N} \omega_i \bar{x}_i \leq c, 0 \leq \bar{x}_i \leq 1 \text{ for } i \in N\}$$

$$\pi_j^4 := \max\{\sum_{i \in N} p_{ij}\bar{x}_i : \sum_{i \in N} \omega_i \bar{x}_i \leq c, \bar{x}_i \in \{0,1\} \text{ for } i \in N\}$$

All of these bounds are valid; however, none of them have polynomial time bounds. Several improvements have been introduced, such as allowing weaker bounds to be used by solving the continuous relaxation, and then forcing $\bar{x}_j = 1$ when computing π_j [11]. We can easily verify that U_{GHS}^4 is the tightest of the bounds, and Gallo et. al. found that the weaker bound of U_{GHS}^3 gave the best trade-off between solution tightness and computational effort in a branch-and-bound algorithm among the bounds [18].

2.4.2 Upper bounds from Chaillou et al.

Chaillou et al. utilize the Lagrangian relaxation for the capacity constraint in QKP with $\lambda \geq 0$ [12]. The problem is rewritten as:

$$\max \sum_{i \in N} \sum_{j \in N} \pi_j x_j - \lambda(\sum_{j \in N} \omega_j x_j - c)$$

$$s.t. x_j \in \{0,1\}, j \in N$$

If we set $\hat{p}_{ij} = p_{ij}$ if $i \neq j$ and $\hat{p}_{ij} = p_{ij} - \lambda\omega_j$ if $i = j$, the relaxed problem could be reformulated as a QP $L^c(QKP, \lambda)$. The Lagrangian dual problem is:

$$\min_{\lambda \geq 0} L^c(QKP, \lambda)$$

U_{CHM} is the bound corrsponding to optimal solution of this problem. Chaillou et al proved the Lagrangian dual problem is a piecewise linear funtion of λ with at most n linear segments.

2.4.3 Upper bounds from Billionnet and Calmels.

Billionnet and Calmels linearize the quadratic term, and then solve the LP-relaxation of the problem [4]. First, they introduced variables y_{ij} to constrain binary x_i, then they reformulated the ILP model is as follows:

$$\begin{aligned} \max \quad & \sum_{i,j\in N, i<j} 2p_{ij}y_{ij} + \sum_{j\in N} p_{jj}x_j \\ \text{s.t.} \quad & \sum_{j\in N} \omega_j x_j \leq c \\ & y_{ij} \leq x_i, i,j \in N, i<j, \\ & y_{ij} \leq x_j, i,j \in N, i<j, \\ & x_i + x_j \leq 1 + y_{ij}, i,j \in N, i<j, \\ & x_j, y_{ij} \in \{0,1\}, i,j \in N, i<j. \end{aligned} \tag{2}$$

We get the bound U^1_{BC} by relaxing the integrality constraints to $0 \leq x_i \leq 1$ and $y_{ij} \geq 0$. The formulation above is not very tight, so they also consider adding n new constraints:

$$\sum_{i\in N\ \{j\}} \omega_i x_i x_j \leq (c-\omega_j)x_j, j \in N$$

Furthermore, they derive a Chvatal-Gomory cut $x_i + x_j + x_k - y_{ij} - y_{ik} - y_{ki} \leq 1$ for each of the combinations of indices $i \neq j \neq k$. The bound U^2_{BC} is given by solving the model with the two new kinds of constraints, and it is tighter than U^1_{BC}. However, it is time consuming to derive this improved bound for large-sized instances. To strike a balance, Billionnet and Calmels decided to solve the model in two steps, in which the process is repeated progressively.

2.4.4 Upper bounds from Caprara et al.

From [11], the first bound U^1_{CPT} by Caprara et al could be derived as the optimal value to the problem:

$$\begin{aligned} \max \quad & \sum_{j\in N} \pi^5_j x_j \\ \text{s.t.} \quad & \sum_{j\in N} \omega_j x_j \leq c \\ & x_j \in \{0,1\}, j \in N \end{aligned} \tag{3}$$

where

$$\pi^5_j := p_{jj} + \max\left\{ \sum_{i\in N\setminus\{j\}} p_{ij}\bar{x}_i : \sum_{i\in N\setminus\{j\}} \omega_i \bar{x}_i \leq (c-\omega_j), \bar{x}_i \in \{0,1\} \text{ for } i \in N\setminus\{j\} \right\}$$

Relaxing the integrality constraints gives us bound U^{1*}_{CPT}. Caprara et al set up a skew-symmetric matrix $\Lambda = \{\lambda_{ij}\}$ where $\lambda_{ij} = -\lambda_{ji}$. Let $QKP(\Lambda)$ denote the problem:

$$\begin{aligned} \max \quad & \sum_{i \in N} \sum_{j \in N} (p_{ij} + \lambda_{ij}) x_i x_j \\ \text{s.t.} \quad & \sum_{j \in N} \omega_j x_j \leq c \\ & x_j \in \{0,1\}, j \in N \end{aligned} \tag{4}$$

$U^{1*}_{CPT}(\Lambda)$ is given by solving the problem above. In order to obtain the tightest bound they solved the Lagrangian dual problem:

$$U^2_{CPT} = \min_{\{\lambda_{ij}:\lambda_{ij}=-\lambda_{ji}\}} U^{1*}_{CPT}(\Lambda)$$

The latter problem could be solved through subgradient optimization leading to the bound $\hat{U}^2_{CPT}$ for some near-optimal matrix Λ of Lagrangian multipliers. Caprara et al discussed the relationship between the bounds and provide further details for the mathematically inclined.

2.4.5 Upper bounds from Michelon and Veilleux

Michelon and Veilleux used a similar method to construct QKP bounds [26]. First they consider the formulation:

$$\begin{aligned} \max \quad & \sum_{i \in N} \sum_{j \in N} p_{ij} x_i x_j \\ \text{s.t.} \quad & \sum_{j \in N} \omega_j y_j \leq c \\ & x_j = y_j, j \in N, \\ & x_j, y_j \in \{0,1\}, j \in N. \end{aligned} \tag{5}$$

We denote the problem $L(QKP, \Lambda)$ as the problem resulting from the relaxation of the constraints $x_j = y_j$ for $j \in N$ using multipliers $\lambda_j \in R$:

$$\begin{aligned} \max \quad & \sum_{i \in N} \sum_{j \in N} p_{ij} x_i x_j - \sum_{j \in N} \lambda_j (x_j - y_j) \\ \text{s.t.} \quad & \sum_{j \in N} \omega_j y_j \leq c \\ & x_j, y_j \in \{0,1\}, j \in N \qquad . \end{aligned} \tag{6}$$

For a given vector Λ of Lagrangian multipliers, they get the bound $U^1_{MV}(\Lambda) = L(QKP, \Lambda)$ and tighter bound by solving the Lagrangian dual problem:

$$U^2_{MV} = \min_{\Lambda \in R^n} U^1_{MV}(\Lambda).$$

The authors explicitly discuss the relationship between their bounds and the bounds from Caprara et al.

2.4.6 Upper bounds from Billionnet et al.

"The bound by Billionnet et al. is based on a partitioning of N items into m disjoint classes $\{I_1, ..., I_m\}$ satisfying $\bigcup_{k=1}^{m} I_k = N$ [7]. The main idea in the paper is to use Lagrangian decomposition to split the problem into m independent subproblems. Each subproblem is solved by enumerating all decision variables in class I_k. For a fixed solution vector x_{I_k} the subproblem is an ordinary linear 0–1 Knapsack Problem which may be solved in O(n) time."[29]

The first upper bound could be written as the sum of m subproblems., such that

$$U^1_{BFS}(\Lambda, M) = \sum_{k=1}^{m} L_k(I_k, \Lambda, M)$$

and U^2_{BFS} is found as a solution to the Lagrangian dual problem:

$$U^2_{BFS} = \min_{\Lambda, M} U^1_{BFS}(\Lambda, M)$$

2.4.7 Upper bounds from Helmberg et al.

Helmberg et. al. derived many upper bounds for QKP based on semidefinite programming [20, 21, 22].

2.5 Lower Bounds

Brute force: As a quick comment on enumeration, if there are n decision variables, then there are at most 2^n subsets of variables that form "legal candidate sets", which leads to exponential complexity $\lambda(2^n)$ where the lambda represents the complexity of the value evaluation (i.e. evaluating the cost of the subset).

2.5.1 Greedy Heuristic

The classic Greedy Heuristic given by Dantzig in [14] is readily adapted for QKP lower bounds as follows:

1. Sample k (say $k = \lfloor N/2 \rfloor$) items from the set of N items.
2. Obtain a set of all pairs from the k items.
3. Sort the items non-increasingly according to the efficiency function $\frac{p_{ij}}{w_i + w_j}$
4. Add into knapsack the pair of items with the highest (efficiency) score, ensuring that the accumulated weight does not exceed the maximum capacity.
5. Repeat steps 1 through 4 until pairs can no longer be added.
6. Fill remaining capacity with singleton items, using the previous greedy approach from Dantzig.

From [31], see notebook here.

2.5.2 Linearizations

These method is important for use in commercial solvers. Two are given, denoted LP1 and LP2 (LP2 called Glover's linearization)

$$\begin{aligned}
\max \quad & \sum_{i \in n} p_i x_i + \sum_{i=1}^{n} \left(\sum_{j=1, i \neq j}^{n} (P_{ij} + P_{ji}) z_{ij} \right) \\
\text{s.t.} \quad & z_{ij} \leq x_i \quad \forall (i,j), i < j \\
& z_{ij} \leq x_i \quad \forall (i,j), i < j \\
& x_i + x_j - 1 \leq z_{ij} \quad \forall (i,j), i < j \\
& z_{ij} \geq 0 \quad \forall (i,j), i < j \\
& x \in \{0,1\}^n
\end{aligned} \tag{LP1}$$

This reformulation uses $\binom{n}{2}$ new objective variables and $3\binom{n}{2}$ new constraints. On the other hand, the Glover's lineraization, given as

$$\begin{aligned}
\max \quad & \sum_{i=1}^{n} p_i x_i + \sum_{i=1}^{n} z_i \\
\text{s.t.} \quad & L_i x_i \leq z_i \leq U_i x_i \quad i = \{1, \ldots, n\} \\
& \sum_{j=1, i \neq j} P_{ij} x_j - U_i (1 - x_i) \leq z_i \quad i = \{1, \ldots, n\} \\
& \sum_{j=1, i \neq j} P_{ij} x_j - L_i (1 - x_i) \geq z_i \quad i = \{1, \ldots, n\} \\
& x \in \{0,1\}^n
\end{aligned} \tag{LP2}$$

where L_i and U_i are lower and upper bounds on $\sum_{j=1, i \neq j}^{n} P_{ij} x_j$. Since we've replaced the expression $\sum_{j=1, i \neq j}^{n} P_{ij} x_i x_j$ with continuous z_i, we can similarly use standard commercial solvers for the problem now. This linearization includes only n auxillary variables and $2n$ constraints to achieve linearity, a huge complexity difference from (LP1).

Bhardwaj [3] derived linearizations which are a modified version of the McCormick linearizations (the first linearization) with only half the number of constraints. The first reformulation linearizes each bilinear term using four inequalities. However, only two of them suffice for each bilinear term. The SOS linearizations Bhardwaj developed would provide a way to distinguish those sufficient inequalities. The two efficient inequalities depend only on the sign of the coefficient of the bilinear term.

2.5.3 Convex Reformulation

Billionnet et. al. developed a Quadratic Convex Reformulation [5, 6]. The objective function (QKP) can be written as

$$x^\top P x$$

using matrix algebra notation. We need to make P PSD for the program to be convex. We modify to

$$x^\top P x + \sum_{i=1}^{n} \left(\sum_{j=1, j\neq i} |P_{ij}| \right) (x_i^2 - x_i) \tag{CQP}$$

where P is thus a diagonally dominant matrix and PSD. This can be solved using commercial mixed integer quadratic packages. Galli and Letchford addressed the importance of this method, and how it could be extended under certain conditions [17]. In most cases, they obtained convex formulations whose lower bounds are at least as good as the semidefinite programming bounds, and sometimes even better. This formulation is discussed in our results for its importance in our empirical results.

2.5.4 Dynamic Programming

The DP formulation is given as such: For each $\gamma \leq c$, define the stage cost $g(m, \gamma)$ to be the value of the most profitable packing of the first m items found with a total weight of γ. That is, let

$$g(m, \gamma) = \max\{\sum_{i\in N} \sum_{j\in N} p_{ij} x_i x_j : \sum_{j\in N} x_j = \gamma, 1 \leq i \leq m\} \tag{7}$$

With an initial state of $g(0, 0)$, a forward DP algorithm gives a suboptimal solution $g(m, \gamma)$ in at most $|N|$ stages (since all weights are positive, no cycles occur).

The DP is essentially the greedy algorithm given above, that considers all possible items in the first stage, and hence adds additional complexity to the first stage. From then on, it proceeds as the greedy algorithm since all return information is known (deterministic shortest path with a finite number of stages).

However, adding stochasticity to the formulation at each stage (i.e. every time we "evaluate" a return, it changes at the next stage, for each asset) or considering the infinite horizon case (which could be justified by considering investors who plan to invest long-term, and change their investment strategy continuously, depending on realizations of asset returns over time) would make the DP formulation stronger than the Greedy Heuristic.

The following note is given on the DP formulation: "While dynamic programming can generate optimal solutions to (linear) knapsack problems, dynamic programming approaches for QKP can only yield a relatively good quality solution, which can serve as a lower bound to the optimal objectives. While it runs in pseudo-polynomial time, it has a large memory requirement [16]."

2.6 Integer Programming Approaches

2.6.1 Valid inequalities

Several valid inequalities can be derived by considering the classical KP polyhedron, the QKP or the QP polyhedron. The inequalities presented in this section are in their basic form, but standard lifting techniques as presented by Padberg can be used [27].

All the following inequalities can, in principle, be used to tighten the bounds presented in §2.4, Upper Bounds. While the upper bound is based on decomposition to some specific subproblems, the valid inequalities will typically be transferred to the formulation of the subproblems.
The 0-1 knapsack polyhedron P is given by:

$$\mathscr{H} = \text{conv}\left\{ x \in \{0,1\}^n : \sum_{j \in N} w_j x_j \leq c \right\} \tag{8}$$

Then the simplest valid inequalities are *cardinality constraints*, first proposed in [2]. Simply, for any set $S \subseteq N$ such that $\sum_{j \in S} w_j > c$ then the inequality

$$\sum_{j \in S} x_j \leq |S| - 1 \tag{9}$$

is valid for $\mathscr{H}$, and is used in upper bound calculations often (specifically eq (5) of [29] for U^2_{GHS}).

Next, weight inequalities were proposed by Weismantel [33]. Here, we assume the subset $T \subseteq N$ satisfies $\sum_{j \in T} w_j < c$ and define the residual capacity as $r = c - \sum_{j \in T} w_j$. Then the *weight inequality* (with respect to T) is defined by

$$\sum_{j \in T} w_j x_j + \sum_{j \in N \setminus T} \max\{0, w_j - r\} x_j \leq \sum_{j \in T} w_j \tag{10}$$

The author proved this inequality is valid for $\mathscr{H}$. These weight inequalities can be extended for disjoint subsets $T_1, T_2, \ldots, T_n \subseteq N$ with weight, individually, less than the knapsack capacity. This is the basis of lifting techniques to define new QKP polytopes, but is beyond the scope of our current study (we may use extensions in computational experiments).

2.6.2 Cover Inequalities

Cover inequalities give one of the most important classes of valid inequalities for knapsack polyhedron. A "cover" of the knapsack polyhedron is defined as a set $C \subseteq N$ where $\sum_{I \in C} \omega_i x_i > c, I \in [n]$. A cover is *minimal* if it does not have a proper subset which is also a cover. The corresponding minimal cover inequality is the weight inequality (9) applied to the minimal cover:

$$\sum_{j \in C} x_j \leq |C| - 1 \tag{11}$$

Cover inequalities are clearly valid for knapsack polyhedron and the set of box constraints and all minimal cover inequalities provides an integer formulation for the knapsack set. One is able to derive faces of the knapsack polyhedron by analyzing cover inequalities of it, which is shown by Padberg [27].

To strengthen the integer formulation of the polyhedron we can introduce *strong* covers. Suppose a cover C = $\{i_1, ..., i_k\}$, where $i_1 < i_2 < ... < i_k$, is "strong" if C is a minimal cover and

$$\sum_{j=1}^{k-1} a_{i_j} + a_p \leq c \tag{SCD}$$

where p $\in [i_k - 1] \backslash C$. Moveover, the extension of a cover is the set $E(C) := C \bigcup \{i_k + 1, ..., n\}$. If the weight ω_i are sorted non-decreasingly, the extension inequalities is valid for P:

$$\sum_{j=1}^{k} x_{i_j} + \sum_{t=i_k+1}^{n} x_t \leq |C| - 1 \tag{EI}$$

A theorem proved by Balas, given below, could provide a complete characterization of the cases in which extension inequalities define facet of P [2].

Theorem 1. *The inequality $x(E) < k - 1$ defines a facet of P if and only if it is the extension inequality of a strong cover C and*

$$\omega(C) - \omega_{i_k} - \omega_{i_{k-1}} + \omega_n \leq c$$

2.6.3 Lifting

Lifting is an important concept to strengthen valid inequalities for the knapsack polyhedron. It takes an inequality $\sum_{i \in S} \omega_i x_i \leq \gamma$ with $S \subseteq N$ that is valid for the restricted knapsack set $K_S := \{x \in K : x_i = 0, i \in N \setminus S\}$ and turns it into a valid inequality for K by lifting coefficients for the variables from 0 to appropriate lifting coefficients. It is also possible to decrease coefficients of variables that are contained in the initial inequality by "down-lifting". The goal, then, is to obtain a stronger or even facet defining inequality for P.

There are two ways to compute the lifting coefficients:

1. Sequential Lifting: Iteratively, one coefficient is computed in each step
2. Simultaneous Lifting: All coefficients computed in one step

One important class of valid inequalities for the knapsack polyhedron is the lifted cover inequalities. Balas and Wolsey, independently, proved that for any minimal cover $C \subseteq N$, there exists one facet-defining lifted cover inequality in the form:

$$\sum_{i \in C} x_i + \sum_{i \in N \backslash C} \alpha_i x_i \leq |C| - 1$$

with $\alpha_i \geq 0$ for all $i \in N \backslash C$ [2, 35].

Given a cover C, classical lifting procedures strengthen the cover inequality by increasing 0-coefficients of variables that are not contained in C. Another way of inequality strengthening, called Coefficient Increased Cover Inequalities, is described by Dietrich and Escudero [15]. Their method also allows the increase of the coefficients of cover variables; their strengthening procedure runs in O(nlog(n)) time .

2.6.4 Pack Inequalities & Cutting Planes

We treat these briefly and reference the authors.

Based on an idea from Weismantel [33], Atamtürk considered pack inequalities [1]. A set $P' \subseteq N$ is a pack if $\omega(P') \leq c$.Then the pack inequality:

$$\sum_{j \in P'} \omega_j x_j \leq \omega(P')$$

is trivially valid. It is dominated by the upper bounds $x_j \leq 1$. However, lifted pack inequalities (LPIs) are not necessarily dominated.

Glover et al chose to use the Gomory cuts

$$\sum_{j \in J} \lfloor \lambda_0 \omega_j + \lambda_j \rfloor x_j + \sum_{j \in [n] \backslash J} \lfloor \lambda_0 \omega_j \rfloor x_j \leq \lfloor \lambda_0 c + \sum_{j \in J} \lambda_j \rfloor$$

where $\lambda \in R_+^{J \times \{0\}}$, and $J \subseteq N$, as cutting planes. After working on surrogate analysis, they found out that these cutting planes can be separated in polynomial time. Moreover, they described how to strengthen these cutting planes and they conjecture that the strengthening process can not be found by classical lifting procedures [19].

3 Application: The Mean-Variance Problem

Fereshteh et al [32] applied knapsack problem to a portfolio selection model under interval uncertainty. Their paper proposes a method to determine the number of shares for each asset in the context of stock market by converting the model into a parametric linear programming model and designing a discrete firefly algorithm to find the near optional solution in large dimension. Moreover, the study was also implemented for some real data from the US stock exchange and the numerical result have shown the method could yield promising outcomes. We proceed in a similar vein of computational experiments, utilizing techniques learned through our course in integer programming (IP) and comparing their effectiveness

A classic use of the Convex Quadratic Programming (CQP) is in the Markowitz Portfolio problem, first introduced in [25], which trades off expected return vs. risk for investment in a "risky asset". The problem is given as such: Consider a scenario where we would like to choose a portfolio that allocates investments into some combination of p risky assets and 1 risk-free asset, while considering a tradeoff between maximizing the expected return of the portfolio and the risk tolerance of the investor. The Markowitz portfolio selection model [25, 10] is a simple framework to pose such a problem. Let $\xi \in \mathbb{R}^p$ be a random variable of the returns from the p risky assets, and define $\mu = \mathbb{E}_\xi \xi$ and $\Sigma = \mathbb{E}_\xi((\xi - \mu)(\xi - \mu)^\mathsf{T})$. Then one formulation of the problem involves solving a convex quadratic program

$$\min_{x \in \mathbb{R}^p} \left\{ x^\mathsf{T} \Sigma x - \gamma \cdot x^\mathsf{T}(\mu - r\mathbf{1}) \;\middle|\; x \geq 0, \|x\|_1 \leq 1 \right\} \tag{12}$$

where: r is the rate of return for the risk-free asset, $\gamma > 0$ trades-off between the returns and risk of the portfolio, and each entry of the vector x gives the fraction of the portfolio allocated to the p risky assets; hence $1 - \sum_{i=1}^p x_i$ is the fraction of the portfolio allocated to the risk-free asset. Importantly, the L1 constraint allocates our knapsack's budget, while the objective represents minimized risk, with a tradeoff for expected return. *This case is important for its generality*, as the risk vs. return trade-off is principal in optimization theory, especially for economists and those in the management sciences.

The above is the Lagrange relaxation (with parameter γ) for the mean-variance problem. To return to the constrained formulation, we write the above program as

$$\min_{\substack{x \in \mathbb{R}^p_+ \\ \|x\|_1 = 1}} \left\{ x^\mathsf{T} \Sigma x \;\middle|\; x^\mathsf{T}(\mu + r\mathbf{1}) > c \right\} \tag{13}$$

and have our normal mean-variance problem of minimizing (investment) risk subject to a guaranteed return of at least c, over a risky asset with (expected) return μ and risk-free asset with guaranteed return r. We can rewrite this as

$$\max_{\substack{x \in \mathbb{R}^p_+ \\ \|x\|_1 = 1}} \left\{ -x^\mathsf{T} \Sigma x \;\middle|\; -x^\mathsf{T}(r\mathbf{1} + \mu) \leq -c \right\} \tag{14}$$

and introduce (dummy) binary variables with appropriate constraints as

$$\begin{aligned} \max_{\substack{x \in \mathbb{R}^p_+ \\ y \in \{0,1\}^p}} \quad & -x^\mathsf{T} \Sigma x \\ \text{s.t.} \quad & -(x \circ y)^\mathsf{T} \cdot (r\mathbf{1} + \mu) \leq -c && \text{(i)} \\ & x_i \leq y_i \quad i \in [p] && \text{(ii)} \\ & \|x\|_1 = 1 && \text{(iii)} \end{aligned} \tag{15}$$

where "$\circ$ represents the Hadamard product, i.e. the componentwise multiplication of vectors x and y. With this addition to the constraint, we have an MIP of the form (QKP), with additional linear constraints (note: all x_i are positive, the L1 constraint can be written as a sum of x components, i.e. $\sum x_i = 1$). Here constraint (i) is our knapsack budget, constraint (ii) selects assets to invest in, and constraint (iii) tells us to invest the full portfolio value, by percentage. We note that the original Markowitz formulation has equality for the constraint (i), however the equality of constraint (iii) ensures we must invest. Studying this problem contributes to a well-studied problem from the integer and combinatorial programming standpoint, as we will apply and interpret bounds and valid inequalities for this domain.

4 Results

In Pisinger's survey, they provide computational results where they consider classical randomly generated QKP instances as proposed by Gallo et al.[18] [29], since these have become the standard tests QKP algorithms. The instances are constructed as follows: let Δ be the *density* of the instance, i.e., the percentage of nonzero elements in the profit matrix P. Each weight w_j is randomly distributed in $[1, 50]$ while the profits $p_{ij} = p_{ji}$ are nonzero with probability Δ, and in this case randomly distributed in $[1, 100]$. Finally, the capacity c is randomly distributed in $[50, \sum_{j=1}^{n} w_j]$.

4.1 Contributions

(1) We studied a new QKP (the mean-variance problem (14)), and give the first MIP formulation (that the authors know of) for the problem in (15). This formulation may be important for large instances of these problems (see §5).

(2) We give performance results (similar to section 8 of the Pisinger survey [29]), for the Quadratic Convex Program (QCP) of the mean-variance problem, see §2.6, item 3. We also (independently) implemented the QKP greedy algorithm and the Lagrangian relaxation of the QCP (relaxing the knapsack constraint). After solving the (exact) QCP given in (15), we measured the performance improvement (by time and objective value) for uses of:

a). Two Reformulations

[1] Lagrangian Relaxation given in (14).

[2] The Convex constrained program of (15).

4.2 Notebook

Please see Jupyter notebook here for results and code. Feel free to contact the authors by e-mail for more information.

5 Discussion & Future Work

It is well known that QKP has numerous practical applications, making it very important to develop both exact algorithms and heuristics for solving large instances of the QKP. In practice, instances with large amounts of data and variables are very common. Recent large QKP instances solved by Caprara et. al. [11], with up to 400 variables, lag far behind the work done for the linear Knapsack Problem, which has empirical results for up to several thousands variables , making tighter and faster bounds needed.

After many trials, researchers found that they could use the tight but computationally expensive bounds to reduce the size of an instance, while computationally cheaper bounds can be used inside a branch-and-bound algorithm. According to the work of the current authors, this is the current best-approach; however, considerations for improvement can be enumerated. For example, one consideration for extension is the introduction of sparsity measures for the QKP (i.e. "density of the QK"). Currently, sparse instance of the QKP are only reported to have been solved for models with less variables than leading dense ones. The authors mention here that the L1 norm used in our formulation is known for its use in *model selection*, and thus often drives variable coefficients to zero for instances with a large number of variables (of course the "zero norm" accomplishes this more explicitly, but does not have all the properties of a "norm").

Moreover, many authors who appeared in our report performed their computational experiments on randomly generated instances with the profits and weights uniformly distributed in a narrow interval. Necessarily, real-world data occur in larger and more varied intervals, which may lead researchers to extend experiments to other solution methods, such as geometric methods, solving correlated problems and subproblems, and graphical approaches.

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